Research Paper Domain Classification System

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Scope: Covers the end-to-end research paper domain classification workflow: PDF input and extraction, shared frozen SPECTER2 embedding generation, concurrent XGBoost/MLP/Logistic Regression classification, output synchronization and comparison, consensus or human review, final domain output, result storage, feedback, and retraining readiness.

Diagram key

☐ Process

◆ Decision

Start Paper Classification
Input Research Paper PDF
Paper Ingestion &amp; Extraction ETL with Airflow or Prefect Text and metadata extraction
Validate Extracted Content
Extraction Successful?
No
Yes
Log Extraction Error
SPECTER2 Embedding Service Hosted frozen transformer Frozen/pre-trained No fine-tuning during classifier updates
Yes
Retry Extraction?
No
Terminate Extraction Failure
Generate Numerical Vector Representation Single shared SPECTER2 embedding
Parallel Classification: Same SPECTER2 Embedding All three execute independently and concurrently
Logistic Regression Classifier (LogReg)
Multi-Layer Perceptron (MLP) MLP Classifier
XGBoost Classifier
Generate LogReg Domains + Confidence
Generate MLP Domains + Confidence
Generate XGBoost Domains + Confidence
Wait for All Three Outputs
Compare Classification Results Predicted domains, confidence scores, agreement/disagreement Accuracy, precision, recall, F1-score Confidence Calibration: temperature scaling and held-out validation
Reliable Classification Consensus?
Yes
No
Human Review Dashboard Inspect low-confidence predictions; correct and label
Select Final Domain Classification Based on comparison results and evaluation criteria
Record Human Correction
Final Research Domain Classification Domain Classification Output Selected domain(s) and calibrated confidence score(s) Structured results
Store Results and Feedback Data Storage: final classification, model outputs, confidence scores, metadata Log classifier performance and disagreements
Update Labeled Dataset Corrections to labeled dataset Available for future classifier retraining
Classification Complete

Research Paper Classification Workflow

Diagram information

This diagram represents an end-to-end workflow for classifying research papers into domains. It ingests and validates a PDF, extracts text and metadata, generates a shared SPECTER2 embedding, and sends it concurrently to XGBoost, MLP, and Logistic Regression classifiers. The system compares predictions and calibrated confidence scores, selects reliable consensus results or routes uncertain cases for human review, then stores outputs and updates the labeled dataset for future retraining.

SOP for Automated Paper Domain Classification

1. Start the paper classification process.
2. Input the research paper PDF.
3. Ingest the paper using an ETL workflow orchestrated by Airflow or Prefect.
   3.1. Extract the paper text.
   3.2. Extract relevant paper metadata.
4. Validate the extracted text and metadata.
5. Determine whether extraction was successful.
   5.1. If extraction fails, log the extraction error.
   5.2. Decide whether extraction should be retried.
   5.3. If retrying is approved, return to the ingestion and extraction step.
   5.4. If retrying is not approved, terminate the extraction failure workflow and record the classification process as complete.
6. If extraction succeeds, submit the extracted content to the SPECTER2 embedding service.
   6.1. Use the hosted frozen, pre-trained transformer.
   6.2. Do not fine-tune SPECTER2 during classifier updates.
7. Generate one shared numerical SPECTER2 vector representation for the paper.
8. Run the three classifiers independently and concurrently using the shared embedding.
   8.1. Execute the XGBoost classifier.
   8.2. Execute the Multi-Layer Perceptron classifier.
   8.3. Execute the Logistic Regression classifier.
9. Generate predicted domains and confidence scores from XGBoost.
10. Generate predicted domains and confidence scores from the MLP classifier.
11. Generate predicted domains and confidence scores from Logistic Regression.
12. Wait until outputs from all three classifiers are available.
13. Compare the classification results.
    13.1. Compare predicted domains.
    13.2. Compare confidence scores and classifier agreement or disagreement.
    13.3. Evaluate accuracy, precision, recall, and F1-score.
    13.4. Apply confidence calibration using temperature scaling and held-out validation results.
14. Determine whether the classification has a reliable consensus.
15. If a reliable consensus exists, select the final domain classification according to the comparison results and evaluation criteria.
16. Produce the final research domain classification output.
    16.1. Record the selected domain or domains.
    16.2. Record the calibrated confidence score or scores.
    16.3. Format the result as structured output.
17. If no reliable consensus exists, route the prediction to the human review dashboard.
    17.1. Inspect low-confidence predictions.
    17.2. Review classifier disagreements.
    17.3. Correct the domain classification and apply the appropriate label.
18. Record the human correction.
19. Store the final classification, model outputs, confidence scores, metadata, and human feedback.
20. Log classifier performance and disagreements.
21. Update the labeled dataset with validated human corrections.
    21.1. Make corrected labels available for future classifier retraining.
22. Mark the classification workflow as complete.